

URUP, Russian Federation

WMO: 321860

Lat: 46.200N

Lon: 150.500E

Elev: 249

StdP: 14.56

Time Zone: 10.00 (E10)

Period: 86-97

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.1	14.3	4.2	6.9	14.6	6.8	7.9	15.8	53.9	26.5	49.9	25.2	21.4	320	0.957

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.0	63.0	58.9	59.9	57.2	57.5	55.3	60.4	61.5	57.9	59.2	55.7	56.9	11.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.9	77.9	61.1	57.4	70.9	58.4	55.1	65.3	56.2	26.9	61.6	25.2	59.3	23.8	56.9	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
45.8	39.8	34.9	11.5	69.2	3.5	3.0	8.9	71.4	6.9	73.1	4.9	74.8	2.3	76.9
			10.1	65.5	3.2	3.3	7.7	67.8	5.8	69.8	4.0	71.6	1.7	74.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.8	24.2	22.6	26.4	32.4	37.4	41.0	46.1	52.0	49.7	44.3	35.0	28.9
	DBStd	10.49	5.01	5.29	3.89	3.36	3.74	3.55	3.95	4.76	3.70	3.73	4.85	4.54
	HDD50	4985	799	767	730	529	390	270	133	34	48	181	451	653
	HDD65	10306	1264	1187	1195	979	855	719	585	404	458	641	901	1118
	CDD50	154	0	0	0	0	0	1	13	95	40	5	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	2	0	0	0	0	0	0	0	2	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.3	21.5	20.7	19.8	18.4	15.4	11.9	11.4	11.9	12.5	15.1	17.8	19.9
Precipitation	PrecAvg	53.7	3.7	3.2	3.7	3.9	3.8	3.4	4.4	5.2	6.2	5.8	5.6	4.8
	PrecMax	67.2	5.7	8.0	7.0	7.1	5.8	6.1	8.9	12.0	10.0	9.2	11.7	7.5
	PrecMin	41.4	2.0	1.5	1.8	1.8	1.6	1.3	1.4	1.5	2.8	2.7	2.6	3.0
	PrecStd	6.6	1.1	1.3	1.2	1.3	1.0	1.1	1.7	2.3	1.8	1.6	1.9	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.6	35.0	36.9	44.8	56.7	58.5	62.0	69.5	64.5	60.1	48.3	43.5
		MCWB	34.3	33.4	34.2	38.5	45.2	51.5	57.3	64.9	58.7	53.3	45.5	41.3
	2%	DB	34.4	33.2	34.2	41.0	49.7	51.5	57.7	64.3	61.0	54.1	45.8	40.2
		MCWB	33.3	32.0	33.0	36.8	43.8	47.4	55.5	61.1	57.6	49.5	44.4	38.7
	5%	DB	33.2	31.9	32.8	38.8	45.6	48.5	54.9	61.5	57.7	51.4	43.9	37.8
		MCWB	32.1	30.5	31.7	36.3	41.4	45.9	52.9	59.4	54.0	48.1	42.6	36.8
	10%	DB	31.8	30.6	31.7	37.0	42.6	46.3	52.2	58.8	55.5	49.6	42.0	35.5
		MCWB	30.6	29.1	30.6	35.3	40.2	44.8	50.9	57.5	53.2	47.3	40.5	33.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.8	34.7	35.6	40.4	47.7	52.4	59.3	66.1	61.2	54.6	46.6	42.2
		MCDB	35.0	35.1	36.2	42.0	53.9	55.0	60.7	68.1	62.2	58.6	47.2	42.3
	2%	WB	33.4	32.6	33.3	38.4	44.3	48.4	56.1	62.3	57.8	51.0	44.9	39.1
		MCDB	33.9	32.9	33.6	39.8	47.8	50.5	57.5	63.5	60.0	52.8	45.6	39.6
	5%	WB	32.3	30.9	32.1	36.9	42.1	46.3	53.4	59.5	55.3	49.2	43.0	36.9
		MCDB	32.9	31.7	32.6	38.1	44.7	47.6	54.5	60.4	56.4	50.4	43.7	37.9
	10%	WB	30.7	29.2	30.8	35.6	40.5	44.9	51.1	57.6	53.6	47.8	41.0	34.0
		MCDB	31.7	30.5	31.5	36.7	42.4	46.2	52.0	58.7	54.8	49.2	41.8	35.3
Mean Daily Temperature Range	5% DB	MDBR	5.6	5.6	5.5	5.8	7.6	7.3	7.5	9.0	8.2	7.4	6.8	6.1
		MCDBR	6.2	6.7	7.2	9.9	14.6	12.4	12.8	13.9	13.1	10.3	9.9	9.3
		MCWBR	6.5	7.3	7.4	7.4	9.7	9.4	10.2	11.3	9.8	8.0	9.6	10.1
	5% WB	MCDBR	6.2	7.3	6.7	8.8	12.2	10.5	12.5	12.5	11.5	9.3	9.7	8.8
		MCWBR	6.6	7.9	7.3	7.1	9.1	8.8	10.5	11.1	9.7	8.1	9.9	10.1
Clear-Sky Solar Irradiance	taub	0.354	0.375	0.438	0.488	0.495	0.482	0.458	0.437	0.392	0.399	0.392	0.366	
	taud	2.128	2.088	1.961	1.895	1.961	2.083	2.231	2.305	2.383	2.189	2.106	2.077	
	Ebn at Noon	221	244	244	244	248	252	257	258	260	233	204	198	
	Edh at Noon	30	38	50	58	56	50	43	38	32	34	30	28	
All-Sky Solar Radiation	RadAvg	356	584	954	1226	1330	1362	1339	1162	1062	774	427	303	
	RadStd	21	48	84	130	143	110	126	84	84	53	42	16	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page